



Initializing Interactive Treasure Hunts in Cultural Heritage Sites: An LLM-Based Approach

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Abstract. Cultural heritage sites are increasingly embracing playful approaches such as treasure hunt games to enhance interactive educational experiences, proving effective in engaging diverse audiences, encouraging exploration, and promoting knowledge retention. However, designing such experiences remains a complex task that requires careful integration of narrative, interactivity, and factual accuracy.

In this article, we propose an approach that leverages large language models (LLMs) to automate the initial drafting of treasure hunt games for cultural heritage sites. Our method allows content curators to specify key parameters—such as the target artefacts to be included in the hunt, intended audience, and narrative styles—after which the system generates a structured sketch of the game. We compare two generation strategies: a basic sequential method and a revised approach that incorporates a pre-planning phase. Our evaluation assesses the resulting drafts in terms of their correctness, consistency, and stylistic coherence.

Results suggest that pre-planning improves the quality of the generated content, producing generally more structured and contextually appropriate outputs. Moreover, we describe some of the remaining challenges, such as the need for interactive and validated co-design mechanisms or the introduction of factual accuracy guarantees in the adventures.

Keywords: Large Language Models · Educational escape room · Games at museums · AI-assisted game design · Automated content generation

1 Introduction

In recent years, cultural heritage sites have increasingly adopted mobile technology to enhance their repertoire of interactive educational experiences, complementing traditional museum tools and practices. Among these digital strategies,

gamification has emerged as a powerful way to make cultural learning more engaging, as noted in numerous literature reviews [7, 15, 17–19].

A particularly effective gamified approach is the treasure hunt-style game, typically implemented in the form of apps for mobile devices. These experiences engage users through interactive elements such as geolocated navigation stages, clue-following, and puzzle-solving. They may also incorporate object collection, customised quizzes, and AR-based scanning of QR codes or venue artefacts, showcasing new immersive alternatives to promote problem-solving and deeper engagement with local heritage [3, 14].

Targeting a wide range of audiences, including children, school groups, university students, and general museum visitors, these gamified interventions have been proven to enhance knowledge retention [28], promote collaborative learning [8], and encourage exploration and contextual learning [5], positioning them as effective tools for cultural interpretation [3]. Despite their potential, the design and implementation of these experiences remain challenging, requiring a careful balance and integration of narrative, interactivity and technology.

To ease the design process, generative artificial intelligence models can provide assistance in brainstorming [30] or generating initial versions of these games. This is in line with what is already present within commercial interactive content creation platforms such as *Kahoot!*¹ or *Quizizz AI*², which offer simple mechanisms supported by Large Language Models (LLMs) as a source of inspiration and novel ideas. When developing content for cultural heritage sites, however, it is essential that the material produced remain factually grounded on the venue’s knowledge base, often comprised of numerous texts with information regarding the artefacts of its distinct exhibits. Given the tendency of LLMs to hallucinate, ensuring factual accuracy is paramount, particularly in learning-centred experiences.

In the past, we have developed *Enigmas* [10], treasure hunt type games for museums supported by AR technology, mixing narrative elements with clues, artefact searches, mini-games, and multiple-choice questions to test players on the content of the exhibition. Historically, designing such adventures has been time-consuming, particularly when it comes to formulating initial ideas for the structure of the experience. With this in mind, in this paper we propose an LLM-based approach to automate the production of these initial drafts in a limited subset of *Enigma* games. Our method involves content curators specifying key domain information, such as the artefacts to be included, the target audience, or the desired narrative style, after which the system generates structured data files to bootstrap the game. We compare two approaches: a baseline method using direct and sequential prompting and a revised version incorporating a pre-planning step. We then evaluate their effectiveness in terms of content correctness, stylistic appropriateness, and overall perceived quality.

¹ <https://kahoot.com/>.

² <https://quizizz.com/quizizz-ai>.

2 Related Work

The rise of LLMs has spurred interest in using these AIs for text generation across domains, including video games. Trained on large text corpora, LLMs generate responses based on user prompts, usually accompanied by a set of rules or conversation history. Without prompt-embedded domain knowledge, their responses rely entirely on input and training data, known as a zero-shot paradigm, whereas adding examples enables few-shot reasoning [29]. These models are not free of reasoning and action errors, however, which has given rise to different strategies to improve the generation quality by means of patterns such as chain-of-thought prompting [25] or ReAct (Reasoning and Acting) [27].

In video games, the use of LLMs has been explored for a variety of tasks including narrative creation, interactive stories, and scene generation [4, 16, 20]. This includes the production of games in the form of visual novels with complex branching that propagate over time [22], although some work reports that LLM-generated dialogues are not always perceived as positively as those written by humans [2]. They have also been applied in level design (e.g., Mario Bros [21] or Sokoban levels [23]) or for game environments and layouts [9, 13].

For cultural heritage sites, LLMs have been used for guided tours and personalized narratives rather than traditional games. Helmy et al. [12] combine a 3D scene generation tool (NeRF) with the LLaMa 3 language model to power a virtual tour guide for Egyptian heritage sites. In turn, Trichopoulos et al. [24] use OpenAI’s GPT model to generate personalized narratives and recommendations for museum visitors. These approaches use conversational interfaces to encourage user exploration and engagement.

In contrast to our work, existing studies—including those focused on cultural heritage applications—have not placed particular emphasis on the factual accuracy of the generated content. Moreover, our study differs from the literature in that a treasure hunt consists of several phases of different nature, each of which requires a certain level of internal consistency and factual accuracy. While existing generators create narratives, quests, questions, or environments in isolation, our approach integrates all components into a cohesive, unified narrative.

3 Games Description and Problem Definition

Enigma games are interactive adventures for mobile devices, inspired by escape rooms and treasure hunts and designed to engage players in story-driven experiences at cultural heritage sites such as museums or archaeological sites. Players assume the role of the protagonist, going through a series of interconnected puzzles, riddles, and quests. The primary goal is to motivate visitors to actively explore their surroundings and interact more deeply with the artefacts on display, dynamizing their experience and encouraging reflection on the content.

The gameplay is structured in distinct stages, each presenting a unique challenge, such as locating artefacts, solving puzzles, answering multiple-choice questions, or unlocking digital locks. These phases are sequential, requiring completion of one to proceed to the next. Players interact with the game via touchscreen,

GPS navigation, and augmented reality features, bringing the venue and its institutions to life. The aim is to foster a sense of immersion and active participation, making players feel like an integral part of the adventure. These games have been used at several institutions in Madrid, Spain, such as the National Museum of Natural Sciences, the Lázaro Galdiano Museum, and the García Santesmases Museum of computer science history.

In this work, we focus on three of the thirteen phases currently available in these games, which form the core structure of most Enigma adventures:

- *Narrative phases*: virtual characters interact via dialogues to tell a story or elaborate on concepts and events that take place in the game.
- *Artefact search phases*: players are prompted to explore the area to locate a museum item based on a treasure hunt-style clue.
- *Multiple-choice question phases*: the system presents questions with a correct answer and distractors to assess the player’s understanding.

Typically, these three kinds of phases are introduced in this same order, giving rise to thematic blocks focusing on each of the artefacts to be discovered by the player: (1) A narrative stage introduces or continues the story and motivates the search for an artefact; (2) a search phase prompts the player to reason about the given clue and explore the area to locate the artefact; and (3) a question phase encourages further scrutiny of the work in search of the correct answer.

4 Approach

Our approach starts from a set of treasure-hunt configuration details provided by the game designer or content curator and generates a JSON file with the definition of an Enigma game based on the information provided (see Fig. 1).

4.1 Game Parameters

To build an adventure based on a solid knowledge of the museum’s objects, the first essential step is to list the artefacts the designer wants to include in the game. If the museum already displays panels with textual descriptions of the works, these can be directly used as input. However, depending on the use case, it may be beneficial to enrich this data with other remarks such as information about nearby objects, metadata regarding the artefact’s location, or any relevant facts that could help generate clues in the treasure hunt. An example of an artefact description from our case study is shown below in Sample 1 for the artwork “Apollo in the Forge of Vulcan”.

“Apollo in the Forge of Vulcan”: “Diego Velázquez’s painting *Apollo in the Forge of Vulcan* (c. 1630) depicts the Roman god Vulcan at work in his forge, surrounded by assistants hammering metal in the intense heat. Vulcan, engaged in forging, represents both creation and destruction. Velázquez contrasts the warm glow of fire with cooler tones...” (1)

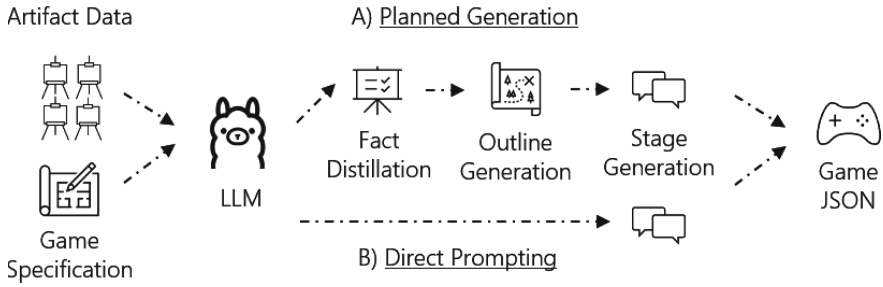


Fig. 1. Treasure Hunt generation flow for both approaches.

Table 1. Sample input for a “Mythology Mystery” specification.

Parameter	Input
General Description	A mythology-based adventure where players uncover the legendary stories hidden in paintings and sculptures
Intention	Provide a deep dive into mythology through artwork
Fact Focus	Myths and legends behind classical artworks
Narrative Style	A mysterious quest where players act as seekers of lost mythological knowledge
Clue Style	Riddles based on myths, e.g., “Seek the god who...”
Question Style	Questions exploring the accuracy of myths in art
Target Audience	Mythology and history enthusiasts
Difficulty	Hard

Once this information is provided, the rest of the parameters start to refer to stylistic and focus considerations both at the overall domain level and for the different phases of the adventure, as can be seen in the example from Table 1. Notably, parameters are provided to control the target audience (e.g., “Mythology and history enthusiasts”), the design intent, the expected difficulty, and a general description of the goal of the application. Numerical inputs to control other aspects of the game, such as the length of narrative blocks or the number of answers per multiple-choice question are also included.

4.2 Fact Distillation and Plan-and-Execute Strategy

Once the designer has entered this information, the game generation process begins. This starts by taking the texts for each of the artefacts and producing a reduced set of succinct facts with the focus indicated by the designer in the **Fact Focus** parameter. Given that the descriptions provided by these institutions can be quite lengthy, the first hypothesis (**H1**) of this study is that distilling the most

relevant facts from these descriptions improves the consistency and accuracy of the clues and questions generated by the LLM. The number of facts output per artefact is controlled by the `Number of Facts` input parameter. We preliminarily argue that this step may contribute to focusing the attention of the model on the pieces of information that the designer considers most pertinent in the given context. An example of the facts suggested by the model for the artwork “Apollo in the Forge of Vulcan” using the sample inputs above is shown below.

1. Velázquez’s painting ‘Apollo in the Forge of Vulcan’ humanizes the Roman god Vulcan, depicting him as a labourer rather than an idealized deity, making it an allegory of craftsmanship, creativity, and transformation.
2. Velázquez masterfully contrasts the warm glow of fire with cooler tones, creating a dramatic atmosphere that blends mythology with naturalism.
3. Velázquez enhances the realism and depth of the scene through his use of light and shadow, showcasing his technical mastery.

During all stages of generation, we make use of Microsoft’s Guidance library [1] to control the output of the LLM in a structured way. This ensures that it is never necessary to perform parsing operations on the data generated by the model and that, in general, there are always mechanisms in place to guarantee the structural integrity of the proposed solutions. For instance, it is possible to restrict the output of the LLM during narrative generations to pick from only valid characters available in the adventure metadata or to impose the generation of an exact number of single-sentence answers in the question stages.

Secondly, given that the generated adventures may become excessively long, this study proposes a second hypothesis (**H2**): that developing an action plan prior to phase generation supports narrative focus and coherence across the game’s stages. To this end, once the most relevant facts of each work have been distilled, we prompt the model to prepare a structured draft in which the points to be followed throughout the adventure are briefly defined. This is in line with common design patterns in prompt engineering, such as Plan-and-Execute, where the model first plans a strategy for solving a complex problem and then executes each of the solving steps in a systematic way.

More specifically, such a plan involves designing a narrative-question-clue sequence for each of the artefacts, along the lines of what was discussed at the end of Sect. 3. For each of the works in the itinerary, the LLM must produce a fact on which to base the clue that leads the player to the piece, the point of emphasis on which to focus in the multiple-choice question to be asked after finding the artefact, and a summary of the narrative from the instant in which the previous object was found to the moment in which the player is to find the next one. Continuing with the example above, the general points generated as a sketch of the artefact block would be:

- **Narrative summary:** “as Kaida and Lyra delve into the world of ancient mythology, they discover that the mysterious collector’s cryptic letter is linked to Velázquez’s enigmatic painting “Apollo in the Forge of Vulcan”, which holds the

key to understanding the transformative power of art and the connection between creativity and craftsmanship, setting the stage for their quest to uncover the secrets hidden within the masterpieces”.

- **Focus of the clue:** “the transformative power of art and the connection between creativity and craftsmanship.”
- **Focus of the question:** “the accuracy of myths in art by asking about the depiction of Vulcan in the painting”.

Once this outline is available, and with it the general narrative, the model makes a proposal of the characters to be used, together with a brief description of each of them. In the ongoing example, proposed characters are **Kaida Asteria**, “a brilliant and enigmatic art historian with a passion for uncovering the hidden secrets of ancient mythologies”, and **Lyra Flynn**, “a charismatic and resourceful adventurer with a knack for deciphering cryptic clues and unravelling mysteries”.

4.3 Stage Generation

Having devised the plan to be followed, our approach then moves on to individually generate the stages of the game according to the general descriptions in the draft. Here, each of the designed stages is created from a sequence of prompts to the LLM in which the style settings, the focus to be followed, and the rules that must be generally respected in any stage of the corresponding type are injected. For instance, in question-type stages, one of the basic restrictions is that answers must not be repeated, while in clue-type stages, a clue must not explicitly reveal the name of the artefact to be searched for.

Structural and design constraints are a key aspect to take into account during the generation process. By the former, we refer to constraints on the shape of the produced content from a formatting point of view:

- A multiple-choice question must have exactly the requested number of possible answers, all of them consisting of a single sentence. The latter is also a restriction in the case of clues.
- A narrative stage can only reference characters previously defined in the game, must not exceed the maximum number of interactions per story block, and each of the interventions must be made up of a single sentence.

By design constraints, we denote rules that cannot generally be verified automatically as they are based on respecting the design intent of the person who specifies the adventure. Some examples for this paper’s use case are:

- A narrative stage should never give overly obvious clues to the next artefact to be encountered in the game (e.g., the name of the artefact itself). Dialogues should not be repeated either directly or paraphrased, must respect the specified storytelling style, and must be cohesive and smoothly link the different stages of the game, especially in terms of providing the necessary context so that subsequent stages are sufficiently grounded.

- A clue stage should not explicitly reveal the name of the artefact to which it leads, but neither should it be so ambiguous or broad that the player has insufficient information to locate it. The level of difficulty, focus, and style should respect the design intentions.
- A multiple-choice question stage should not explicitly reveal the answer in the question, nor list the name of the artefact as one of the possible choices. Responses should not be repeated, either verbatim or in paraphrased but essentially equivalent versions. In addition, the correct answer should not be made more emphatic than the distractors (e.g., with an exclamation mark at the end). The level of difficulty, focus, and style should respect the design intentions. Most importantly, questions must always be correct and respect the factual data provided by the user's descriptions. The latter is particularly important in any adventure with a pedagogical focus.

With this, a final stage sequence for the running example would be:

- **Narrative:**
Kaida: The symbol seems to be a variant of the ancient Greek character for the Muses.
Lyra: That's a great observation, Kaida.
Kaida: I think it's more than just a coincidence.
Lyra: Then we need to find out what Velázquez's paintings have to do with it.
Kaida: The paintings are said to hold the key to understanding the power of art.
- **Clue:** Seek the divine craftsman where fiery passion forges the very essence of art itself.
- **Question:** What is the primary representation of Vulcan in Velázquez's "Apollo in the Forge of Vulcan", according to the mythological context depicted in the artwork?
 - **Correct Answer:** Vulcan is portrayed as a labourer rather than an idealized deity, humanizing him and making the painting an allegory of craftsmanship, creativity, and transformation.
 - **Distractor 1:** The painting shows Vulcan as a powerful, godlike figure, emphasizing his divine nature and mythological significance.
 - **Distractor 2:** The artwork depicts Vulcan as a mere mortal, struggling to control the fiery forge, highlighting his vulnerability and humanity.
 - **Distractor 3:** The painting presents Vulcan as a symbol of destruction, emphasizing his role in the mythological context and the chaos he brings to the world.

While we write these examples here in a simplified form to improve readability, note that the output of the system is delivered as a JSON file containing the definitions of each of the phases, metadata, and character listings. This structure is repeated as many times as artefacts included in the game, with a final additional narrative block between the point at which the player finds the last artefact and the end of the game. All the prompts and Guidance code used for the generation can be found in the project repository³.

³ Repository Link: <https://github.com/pgutierrez858/enigma-llms>.

5 Experiments and Results

As introduced in Sect. 4, we aim to validate two hypotheses: **(H1)** distilling relevant facts about the artefacts to be encountered reduces the number of factual errors and improves the overall consistency of adventures; and **(H2)** applying a Plan-and-Execute strategy prior to the generation of game stages has an overall positive effect on design error rates. We evaluate this by comparing the planned approach with strategies that do not make use of these preliminary steps (direct) on 10 use cases available in the project repository, similar to the one in Table 1. These test cases simulate realistic scenarios for designing an interactive treasure hunt, varying in theme, focus, difficulty, and descriptions of the target audience and styles to be followed. A fixed sequence of 4 popular artworks from the Prado Museum was set as input for all use cases. All the experiments in this section were performed using a Llama 3.1 8B model, queried through Guidance calls, and on a PC with 16 GB of RAM and an Intel Core i7-9750H, 2.60 GHz processor.

For each of the generated stages, we define three possible creation errors: correctness failure (i.e., the phase includes false information), style failure (i.e., the phase does not follow the style stated in the input, or presents some form of generation error), and finally domain failure (i.e., the phase does not conform to the general design constraints). Note that the failure cases do not necessarily have to be mutually exclusive. Table 2 reports the error counts by type found upon close inspection of each of the generated adventures. The complete issue log together with more detailed notes can be found in the project repository.

Table 2. Error Type Counts by Experiment Type and Stage Group.

Stage Type	Correctness		Domain		Style	
	Planned	Direct	Planned	Direct	Planned	Direct
Narrative	0	0	31*	16	12	20
Question	2	4	1	1	2	4
Clue	0	0	0	0	0	0

The first remark to be made here is that while the table includes rows for errors related to clue-type stages, all of their values are 0, as no errors were encountered for any of the categories described in the collection of adventures generated for both modalities. In order to understand the error counts for each of the categories in the table, it is nonetheless necessary to carry out a more exhaustive analysis of the nature of the errors recorded.

In the planned generation scenario, one of the most conspicuous features is the high number of domain errors observed, almost twice as many than with direct generation. On closer inspection, 24 of the 31 reported errors correspond to the problem “The next artefact is revealed in the narrative ahead of time”, meaning that the narrative explicitly mentions the name of the artefact that

the player must find in the next clue-type phase. This is true at some point in a large majority of the adventures produced with this method. The remaining mistakes correspond mostly to generation problems arising from the tendency of the model to repeat phrases or whole blocks of context, effectively invalidating an affected narrative block, or problems related to stopping rules of the regular expressions used to control generation colliding with the intention of the model. As an example, we have the case of sentences terminated prematurely by writing “Mr.”; since the regular expression decides that a valid sentence ends in a full stop, this halts generation, leaving the phrase half-finished. Problems with questions are mostly linked to the introduction of distractors that are correct or whose phrasing makes them too close to being correct. Correctness errors for questions are tied to suggestions of invalid correct answers.

Table 3. Aggregated execution time and word count statistics both approaches.

Category	Planned				Direct			
	Time (s)		Words		Time (s)		Words	
	Mean	Std	Mean	Std	Mean	Std	Mean	Std
Narrative	149.42	55.40	339.87	246.61	140.49	62.46	331.38	294.24
Question	84.84	8.74	99.85	26.50	80.88	7.57	91.13	24.18
Clue	54.68	2.89	24.43	4.80	53.58	2.55	24.45	4.68
Character Generation	44.08	1.48	16.85	3.37	42.40	1.53	16.90	3.63
Fact Extraction	165.03	4.89	—	—	—	—	—	—
Plan Generation	443.45	18.30	—	—	—	—	—	—

In the case of direct generation, a higher number of stylistic errors are observed. These often stem from two issues: the model either fails to follow the designer’s input or, more commonly, gets caught in a loop, repeatedly generating the same sentences. This problem can spread across multiple stages of the narrative—something that occurs far less frequently in the planned approach. Besides this issue, domain errors in narratives of this generation type are typically more disruptive than those of the previous method, as they stem largely from inconsistencies in dialogue or story. For instance, in the use case “Interactive Family Tour”, the characters, both children, hold conversations while referring to each other as “mum” and introduce this into their interactions in a significantly confusing way, producing a difficult-to-understand narrative. On the other hand, several situations were observed in which the model references artworks unrelated to the user’s input as if they were the next artefacts to be found by the player. Something similar happens, for example, in the use case “Time Travel Adventure”, where the model implies repeatedly that the adventure was focused on the works of authors outside the visit, such as Leonardo Da Vinci.

Regarding questions, on top of the problems present in the planned model, errors can arise that are linked to an excess of information both in the question

(e.g., revealing data on the correct answer) and in the correct answers, as these may become overly verbose and clearly different from the distractors in expressive tone. This problem of overly long questions also translates in specific cases into sentences truncated for exceeding the token limit during generation.

The average time to generate each phase type for both approaches, as well as the average word count in phase definitions, are listed in Table 3. The only notable numerical difference is that the planned version takes about 10 min longer on average than the direct one, corresponding to the execution time of the fact extraction and pre-planning part of the approach. However, since neither method needs to operate in real time, this might not prove relevant in practice.

6 Discussion

In Sect. 5, we focused on providing objective data to draw an initial quantitative profile of our approaches. From a practical point of view, however, it is essential to analyse the overall quality of the adventures created using each method. In this sense, we are interested in understanding which adventures best meet the objective of generating a first skeleton for an adventure based on cultural heritage works. This refers to suggestions that a designer or curator would be able to use as a starting point to build upon in the creative process.

With this in mind, there is a clear tendency for planned adventures to be more consistent with respect to a clear narrative scheme. That is, this methodology tends to result in schemes with a more directed and centred narrative focus throughout the different dialogue blocks than its counterpart without planning. Also, domain errors found with planning are generally easier to correct than those introduced in the direct approach, in that they are often reduced to simply revealing the name of the artefact of the next clue; thus, in many cases it is sufficient to omit this information to obtain a valid phase.

On the other hand, it is not uncommon to find minor hallucinations in the narrative discourse of the direct method, where the model occasionally makes comments regarding works that were never entered as input and therefore have no reason to be in the collection. For example, in the use case “Museum Treasure Hunt”,⁴ the characters discuss that the piece “The Ecstasy of St. Teresa” caught their attention, without this being part of the actual collection, and then talk about Caravaggio’s “The Taking of Christ” holding the key to the next clue, again without this being substantiated by any input and therefore being incorrect from a design point of view. While the planned model is not without these flaws (see the same use case where “The Wynter Gallery’s Dutch art gallery” is mentioned), these references tend to be more anecdotal than in its counterpart.

As previously discussed, clues are always valid during generation and can often be considered compelling suggestions, similar to those a designer might propose. However, it is important to note that the clues are generated without the model being aware of all the possible works in the environment where the target

⁴ Full examples may be found in <https://github.com/pgutierrez858/enigma-llms/tree/master/experiments/results>.

artefact is located, and therefore the generation cannot account for possible distractors the riddle could lead to. Now, writing engaging clues for the target audience with a single solution is a challenge and, based on our experience, is something whose validation often depends on user playtesting.

The questions generated by both approaches are generally valid, adhering to the source text, the designer's intention, and the specified difficulty, making them useful first suggestions in the adventure construction process. However, it is important to note that both methods sometimes generate questions that, while correct, are not directly based on the text but likely on the information used to train the original model. While this is not necessarily a flaw, it can lead to hallucinations, for instance, in situations when multiple artefacts seen in training texts share the same name and the model references the wrong one.

7 Conclusions and Future Work

In this paper we introduce a first approach to the automatic generation of treasure hunt games for cultural heritage sites supported by large language models. Starting from a user input specifying the artefacts to be included in the adventure and various textual design parameters, our approach produces a JSON-structured draft of a preliminary quest that strives to comply with the entered constraints. Following the discussion in Sects. 5 and 6, we find that our approach provides a tentative validation of the hypotheses raised in Sect. 4, proposing that the application of a pre-planning strategy has a positive effect on the correctness, consistency, and style of the generated drafts.

There are, however, limitations and points for future work. The first is the potential bias incurred in the selection of the design use cases, which were gathered manually by the authors. The use of a broader dataset derived from interviews with professional content curators in museums is left for future work. Secondly, while here we focus on a pilot for the generation of consistent initial versions of such games, comprehensive usability evaluations, both of how useful the LLM responses are to the designer and a detailed analysis of problems that may arise in the flow of human interaction with the system, remain to be carried out.

Thirdly, our approach as it stands does not allow for interactivity during creation. Alongside the approaches in this work and in the literature, there is a developing interest in the notion of co-design, or turning the human designer into an active actor in the generation algorithm, requesting their insight and participation at different points throughout the applied method [9, 26]. In this sense, one of the clearest lines of future work consists of incorporating this paradigm into the workflow for creating treasure hunts, allowing a scenario in which the designer can make use of these generative tools at any point in the creative process and not only in the bootstrapping phase.

Finally, while more advanced models such as GPT-3.5 or GPT-4 exist that have been found to work well for a multitude of tasks [6], their proprietary nature raises concerns. Llama 3.1 Instruct 8B was used due to it being an open

source model that can be deployed on modest hardware and without the need for a dedicated GPU, while remaining performant for general applications [11]. This is relevant, as not all cultural institutions have the resources to run more demanding models. While this explains the a priori high execution times shown in the results section, it remains for future work to explore more efficient generation strategies, especially if interactivity is to be introduced in the workflow.

References

1. guidance-ai/guidance (2025). <https://github.com/guidance-ai/guidance>. Original-date: 2022-11-10T18:21:45Z
2. Akoury, N., Yang, Q., Iyyer, M.: A framework for exploring player perceptions of LLM-generated dialogue in commercial video games. In: Findings of the Association for Computational Linguistics: EMNLP 2023, pp. 2295–2311. Association for Computational Linguistics, Singapore (2023). <https://doi.org/10.18653/v1/2023.findings-emnlp.151>, <https://aclanthology.org/2023.findings-emnlp.151>
3. Ardito, C., Costabile, M.F., De Angeli, A., Lanzilotti, R.: Enriching archaeological parks with contextual sounds and mobile technology. *ACM Trans. Comput.-Hum. Interact.* **19**(4), 1–30 (12 2012). <https://doi.org/10.1145/2395131.2395136>
4. Buongiorno, S., Klinkert, L., Zhuang, Z., Chawla, T., Clark, C.: PANGeA: procedural artificial narrative using generative AI for turn-based, role- playing video games. In: Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment, vol. 20, no. 1, pp. 156–166 (2024). <https://doi.org/10.1609/aiide.v20i1.31876>, <https://ojs.aaai.org/index.php/AIIDE/article/view/31876>
5. Cesário, V., Radeta, M., Matos, S., Nisi, V.: The ocean game. In: Extended Abstracts Publication of the Annual Symposium on Computer-Human Interaction in Play, pp. 99–109. ACM (2017). <https://doi.org/10.1145/3130859.3131435>
6. Chen, X., et al.: How robust is GPT-3.5 to predecessors? A comprehensive study on language understanding tasks (2023). <https://arxiv.org/abs/2303.00293>
7. DaCosta, B., Kinsell, C.: Serious games in cultural heritage: a review of practices and considerations in the design of location-based games. *Educ. Sci.* **13**(1), 47 (2022). <https://doi.org/10.3390/educsci13010047>, <https://www.mdpi.com/2227-7102/13/1/47>
8. Di Nezza, M., De Santis, A., D’Addezio, G.: Cityquest and ‘Caccia al...Tesoro dei Castelli’. La nuova frontiera della divulgazione formato 2.0. *Rendiconti Online della Società Geologica Italiana* **45**, 17–22 (2018). <https://doi.org/10.3301/rol.2018.23>
9. Gallotta, R., Liapis, A., Yannakakis, G.: Consistent game content creation via function calling for large language models. In: 2024 IEEE Conference on Games (CoG), pp. 1–4. IEEE, Milan (2024). <https://doi.org/10.1109/CoG60054.2024.10645599>, <https://ieeexplore.ieee.org/document/10645599/>
10. Gonzalez-Calero, P., Camps-Ortueta, I., Gutiérrez-Sánchez, P., Gómez-Martín, P.: On the importance of contextualizing an educational escape room activity. *Electron. J. e-Learn.* **22**, 43–56 (2024). <https://doi.org/10.34190/ejel.22.4.3199>
11. Grattafiori, A., et al.: The LLaMA 3 herd of models (2024). <https://arxiv.org/abs/2407.21783>
12. Helmy, M., et al.: Navigating the world with an intelligent tourist guide using generative AI. In: 2024 International Telecommunications Conference (ITC-Egypt), pp. 1–6. IEEE, Cairo (2024). <https://doi.org/10.1109/ITC-Egypt61547.2024.10620592>, <https://ieeexplore.ieee.org/document/10620592/>

13. Hu, S., Huang, Z., Hu, C., Liu, J.: 3D building generation in minecraft via large language models (2024). <https://arxiv.org/abs/2406.08751>
14. Camps-Ortueta, I., Rodríguez-Muñoz, J.M., Gómez-Martín, P.P., González-Calero, P.A.: Combining augmented reality with real maps to promote social interaction in treasure hunts. In: Conference of the Spanish Association for Videogames Sciences (2017)
15. Khan, I., Melro, A., Carla, A., Oliveira, L.: Systematic review on gamification and cultural heritage dissemination. *J. Digit. Media Interact.* **3**(8), 19–41 (2020). <https://doi.org/10.34624/JDMI.V3I8.21934>, <https://proa.ua.pt/index.php/jdmi/article/view/21934>
16. Kumaran, V., Rowe, J., Lester, J.: NarrativeGenie: generating narrative beats and dynamic storytelling with large language models. In: Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment, vol. 20, no. 1, pp. 76–86 (2024). <https://doi.org/10.1609/aiide.v20i1.31868>, <https://ojs.aaai.org/index.php/AIIDE/article/view/31868>
17. Malegiannaki, I., Daradoumis, T.: Analyzing the educational design, use and effect of spatial games for cultural heritage: a literature review. *Comput. Educ.* **108**, 1–10 (2017). <https://doi.org/10.1016/j.compedu.2017.01.007>, <https://linkinghub.elsevier.com/retrieve/pii/S0360131517300076>
18. Mortara, M., Catalano, C.E., Bellotti, F., Fiucci, G., Houry-Panchetti, M., Petridis, P.: Learning cultural heritage by serious games. *J. Cult. Heritage* **15**(3), 318–325 (2014). <https://doi.org/10.1016/j.culher.2013.04.004>, <https://linkinghub.elsevier.com/retrieve/pii/S1296207413001349>
19. Paliokas, I., Sylaiou, S.: The use of serious games in museum visits and exhibitions: a systematic mapping study. In: 2016 8th International Conference on Games and Virtual Worlds for Serious Applications (VS-GAMES), pp. 1–8. IEEE, Barcelona (2016). <https://doi.org/10.1109/VS-GAMES.2016.7590371>, <http://ieeexplore.ieee.org/document/7590371/>
20. Short, A.R.: Designing fictional worlds for play through large language models. In: Volume 3B: 50th Design Automation Conference (DAC), p. V03BT03A051. American Society of Mechanical Engineers, Washington, DC (2024). <https://doi.org/10.1115/DETC2024-143923>, <https://asmedigitalcollection.asme.org/IDETC-CIE/proceedings/IDETC-CIE2024/88377/V03BT03A051/1208864>
21. Sudhakaran, S., González-Duque, M., Glanois, C., Freiberger, M., Najarro, E., Risi, S.: MarioGPT: open-ended Text2Level generation through large language models (2023). <https://doi.org/10.48550/arXiv.2302.05981>, [arXiv:2302.05981](https://arxiv.org/abs/2302.05981)
22. Taveekitworachai, P., Nimpattanavong, C., Gursesli, M.C., Lanata, A., Guazzini, A., Thawonmas, R.: Multiverse of greatness: generating story branches with LLMs (2024). <https://doi.org/10.48550/arXiv.2411.14672>, [arXiv:2411.14672](https://arxiv.org/abs/2411.14672)
23. Todd, G., Earle, S., Nasir, M.U., Green, M.C., Togelius, J.: Level generation through large language models. In: Proceedings of the 18th International Conference on the Foundations of Digital Games, pp. 1–8. ACM, Lisbon (2023). <https://doi.org/10.1145/3582437.3587211>, <https://dl.acm.org/doi/10.1145/3582437.3587211>
24. Trichopoulos, G.: Large language models for cultural heritage. In: Proceedings of the 2nd International Conference of the ACM Greek SIGCHI Chapter, pp. 1–5. ACM, Athens (2023). <https://doi.org/10.1145/3609987.3610018>, <https://dl.acm.org/doi/10.1145/3609987.3610018>
25. Wei, J., et al.: Chain-of-thought prompting elicits reasoning in large language models (2023). <https://doi.org/10.48550/arXiv.2201.11903>, [arXiv:2201.11903](https://arxiv.org/abs/2201.11903)

26. Yannakakis, G.N., Liapis, A., Alexopoulos, C.: Mixed-initiative co-creativity (2014). <https://www.semanticscholar.org/paper/Mixed-initiative-co-creativity-Yannakakis-Liapis/0a410eecf3b23042f95ebfbe0ffdc08ea697c9d9>
27. Yao, S., et al.: ReAct: synergizing reasoning and acting in language models. In: International Conference on Learning Representations (ICLR) (2023). <https://par.nsf.gov/biblio/10451467-react-synergizing-reasoning-acting-language-models>
28. Yu, J., et al.: Personalized treasure hunt game for proactive museum appreciation by analyzing guide app operation log. In: Goh, D.H., Chen, S.J., Tuarob, S. (eds.) ICADL 2023. LNCS, vol. 14458, pp. 30–45. Springer, Singapore (2023). https://doi.org/10.1007/978-981-99-8088-8_3
29. Yu, S., et al.: Few-shot generative conversational query rewriting. In: Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 1933–1936. ACM, Virtual Event China (2020). <https://doi.org/10.1145/3397271.3401323>, <https://dl.acm.org/doi/10.1145/3397271.3401323>
30. Yu-Han, C., Chun-Ching, C.: Investigating the impact of generative artificial intelligence on brainstorming: a preliminary study. In: 2023 International Conference on Consumer Electronics - Taiwan (ICCE-Taiwan), pp. 193–194 (2023). <https://doi.org/10.1109/ICCE-Taiwan58799.2023.10226617>